

Before operating systems were realised on mobile phones, the task of handling the features on a phone was carried out through tedious assembly language programming. Early phones had dot-matrix LCD screens; the number of lines and characters were the specifications by which one screen was graded over the other, and so was the phone in the same respect.

The development on hardware such as advanced microcontroller chips which were not only fast but also supported high-level language (C++ and JAVA) programming and easy interfacing of additional devices such as a camera led to the era of feature-rich mobile telephony. The operating system was a result of these basic developments; mobile applications and games followed with the development of the OS.

The microcontrollers or microprocessors used in a mobile phone need to be programmed to perform their various tasks. This leads us to embedded programming, a technique by which microcontrollers and microprocessors are coded using either low-level languages such as assembly language (family of the 8051 command set) or high-level languages (such as C, C++ and Java). The implementation of high-level languages made it easy to encode programs for complex functionality, which is otherwise a very tedious task using assembly language.

Using high level languages, a programmer can either develop an embedded code for the functioning of the mobile phone, or can develop an OS if the hardware resources (memory capacity and processor) employed on the phone can handle it. High-level languages can also be used to develop applications and games.

This task can also be carried out by good use of Software Development Kits (SDKs). Operating systems such as Symbian OS, Windows Mobile and Linux are battling it out to be the prominent player for mobile phones.